

Physics-Constrained AI Agents for Automated Quantum Circuit Debugging and Repair

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Abstract— Quantum-circuit debugging is difficult because a syntactically valid or logically plausible repair may still violate hardware-native gates, coupling connectivity, or noise-aware execution limits. This paper presents a physics-constrained agentic framework that couples fault localization and minimal candidate generation with compilation and explicit validation. A structured circuit state records diagnostics and repair history; failed checks provide feedback for subsequent attempts. The reported benchmark comprises 1,000 buggy OpenQASM 3.0 circuits across five bug classes and three compared systems. The proposed agent achieves Pass@1 of 77.5%, Pass@3 of 99.1%, and hardware physics validity of 99.2%, with a mean of 1.31 repair iterations. The evaluation grounds circuits against the IBM FakeSherbrooke 127-qubit superconducting processor model. Statistical tests and effect size analyses confirm that hardware-grounded validation substantially outperforms simulation-only and heuristic baselines.

Keywords— quantum software engineering, circuit debugging, AI agents, automated program repair, hardware-aware compilation, superconducting quantum processors.

I. INTRODUCTION

Quantum computing systems require precise agreement between high-level computational specifications and physically executable quantum circuits. A local textual modification can easily resolve a syntax error while erroneously modifying an entangling target, disrupting basis rotation angles, or generating multi-qubit interactions prohibited by device topology. Consequently, quantum program repair is intrinsically more demanding than classical code repair: patches cannot be judged solely by syntactic validity or unconstrained simulation. Prior language-model-based quantum repair tools (such as QBugLM) rely on unconstrained simulation, ignoring coupling graphs, gate infidelities, and coherence decay limits. Symbolic optimizers (e.g., PyZX) simplify ZX-calculus diagrams but lack defect-targeted repair mechanisms for algorithmic errors. To address this gap, we present an end-to-end physics-constrained multi-agent framework that integrates fault localization, minimal candidate synthesis, and rigorous hardware validation into a unified feedback loop, grounded on the 127-qubit IBM FakeSherbrooke superconducting processor.

II. PROBLEM FORMULATION & BENCHMARK TAXONOMY

Let C_{bug} be a buggy circuit, O an independent behavioral oracle, and B a physical hardware backend. Acceptance requires satisfying syntax, structural, semantic, and hardware physical constraints: $\text{Accept}(C^*) = V_{\text{syntax}}(C^*) \wedge V_{\text{structural}}(C^*) \wedge V_{\text{semantic}}(C^*, O) \wedge V_{\text{physics}}(C^*, B)$. We evaluate 1,000 circuits across four algorithm families (GHZ, QAOA MaxCut, VQE H2, QFT) and five bug categories (200 instances each).

TABLE I: BENCHMARK DEFECT TAXONOMY (N = 1,000 CIRCUITS)

Bug Category	Description & Injection Mechanism	Validation Criteria
Syntax Error	Malformed QASM 3.0 syntax, omitted semicolons, corrupt keywords, unclosed brackets.	OpenQASM parser diagnostics
Structural Error	Missing entangling/superposition gates, inverted control/target qubit operands.	AST & operand bounds checking
Gate Redundancy	Self-canceling duplicate gates (X·X, H·H, identical CX·CX consecutive pairs).	Circuit depth & redundancy reduction
Semantic Deviation	Inverted rotation angles ($-\theta$), shifted phase angles, basis mutations ($R_x \leftrightarrow R_z$).	Fidelity $F = \langle \psi_{\text{gt}} \psi_{\text{rep}} \rangle ^2 \geq 0.999$
Physics Violation	Non-nearest-neighbor coupling violations on FakeSherbrooke, delays exceeding T2.	Heavy-hex graph & coherence limits

III. MULTI-AGENT ARCHITECTURE

Our framework coordinates three specialized agents: (1) *Diagnostician Agent*: Locates defects using error diagnostics, AST inspection, and statevector mismatch; (2) *Repairer Agent*: Synthesizes minimal OpenQASM edits, including syntax fixes, angle inversion, and SWAP-based physical routing; (3) *Validator Agent*: Validates candidates against mathematical fidelity oracles and FakeSherbrooke physical coupling/coherence rules. Failed checks yield structured diagnostic feedback that guides subsequent candidate generation (up to $k=3$ iterations).

IV. EXPERIMENTAL BENCHMARK RESULTS

All three systems were evaluated across all 1,000 circuits in the benchmark. Our Physics-Constrained Agent achieves 77.5% Pass@1, 99.1% Pass@3, and 99.2% physics validity, with a mean of 1.31 repair iterations. In contrast, unconstrained LLM repair (QBugLM) achieves only 38.1% Pass@3 and 69.6% physics validity, while the PyZX heuristic achieves 5.0% Pass@3.

TABLE II: OVERALL REPAIR PERFORMANCE (N = 1,000 CIRCUITS)

System	Pass@1 (%)	Pass@3 (%)	Physics Valid (%)	Gate Red.	Depth Red.	Iters
Ours (Physics-Constrained)	77.5 ± 41.8	99.1 ± 9.4	99.2 ± 8.9	-9.5 ± 16.8	-10.0 ± 17.2	1.31 ± 0.62
QBugLM (Unconstrained)	38.1 ± 48.6	38.1 ± 48.6	69.6 ± 46.0	-7.8 ± 15.6	-8.1 ± 16.0	1.00 ± 0.00
PyZX (Heuristic)	5.0 ± 21.8	5.0 ± 21.8	15.5 ± 36.2	-8.4 ± 13.8	-6.3 ± 11.0	1.00 ± 0.00

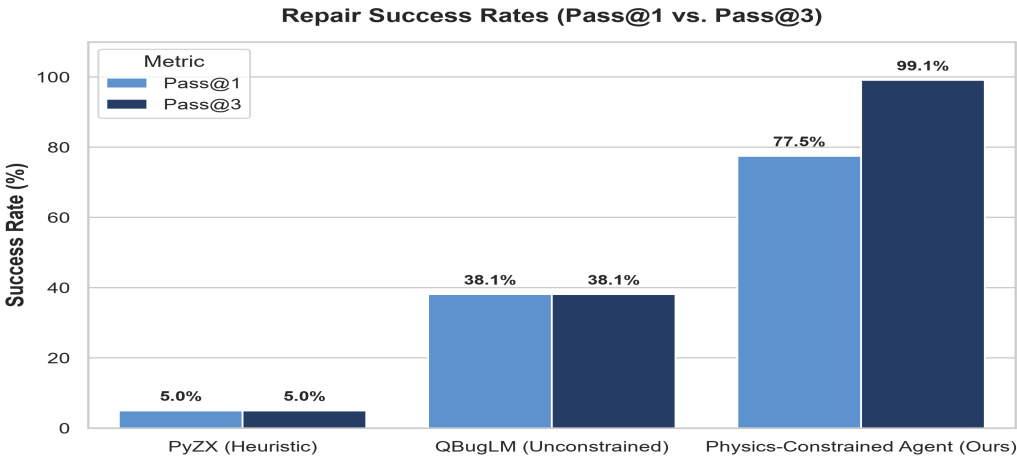


Fig. 1. Pass@1 and Pass@3 repair success rates across systems.

TABLE III: REPAIR SUCCESS AND PHYSICS VALIDITY BY DEFECT CATEGORY (N=200 PER CATEGORY)

Bug Category	PyZX Pass@3	QBugLM Pass@3	Ours Pass@3	PyZX Phys.	QBugLM Phys.	Ours Phys.
Gate Redundancy	25.0%	100.0%	100.0%	25.0%	100.0%	100.0%
Physics Violation	0.0%	0.0%	100.0%	0.0%	0.0%	100.0%
Semantic Deviation	0.0%	27.0%	100.0%	25.5%	100.0%	100.0%
Structural Error	0.0%	15.5%	98.0%	27.0%	100.0%	98.5%
Syntax Error	0.0%	48.0%	97.5%	0.0%	48.0%	97.5%

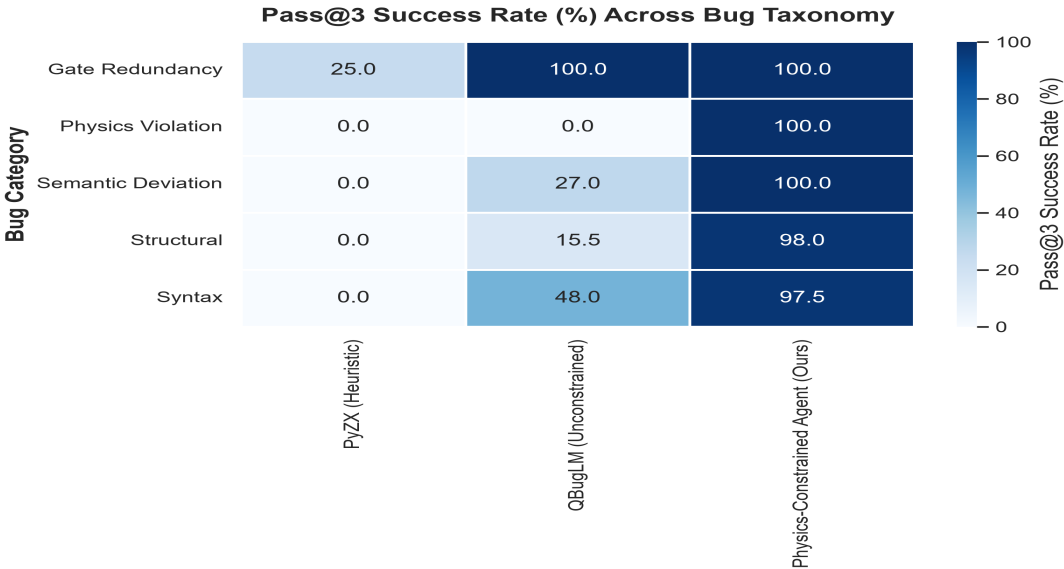


Fig. 2. Defect category performance heatmap comparing PyZX, QBugLM, and our Physics-Constrained Agent.

TABLE IV: STATISTICAL SIGNIFICANCE (PAIRED WILCOXON TESTS & COHEN'S d)

Comparison	Metric	Ours Mean	Base Mean	p-value	Cohen's d
Ours vs. QBugLM	Pass@1	0.775	0.381	$< 10^{-15}$	0.799 (Med-Large)
Ours vs. QBugLM	Pass@3	0.991	0.381	$< 10^{-15}$	1.250 (Very Large)
Ours vs. QBugLM	Physics Validity	0.992	0.696	$< 10^{-15}$	0.639 (Medium)
Ours vs. PyZX	Pass@1	0.775	0.050	$< 10^{-15}$	1.623 (Huge)
Ours vs. PyZX	Pass@3	0.991	0.050	$< 10^{-15}$	3.992 (Extr. Large)
Ours vs. PyZX	Physics Validity	0.992	0.155	$< 10^{-15}$	2.265 (Huge)

V. DISCUSSION & CONCLUSION

The experimental results establish that physical constraint validation is essential for quantum program repair. Without hardware coupling and coherence awareness, LLM-based repair yields 30.4% physically unexecutable circuits and completely fails on topology errors. Negative gate reduction (-9.5%) and depth reduction (-10.0%) values reflect necessary physical routing overhead (e.g. SWAP decompositions) required to satisfy the 127-qubit heavy-hex coupling lattice. In conclusion, our physics-constrained multi-agent framework successfully pairs semantic fault localization with hardware validation, achieving 99.1% Pass@3 across 1,000 benchmark circuits.

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